

Test Case ID: NT-TC-01

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): User Sign-Up / Registration

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description: This test case verifies that a new user can successfully create an account on the NutriTrack application using a valid email and password on an iOS device.

Context: User registration is required to allow new users to create a personal profile and begin tracking their nutrition data.

Precondition

- The NutriTrack app is installed on an iOS device
- User does not already have an existing account
- A stable internet connection is available

Dependency

- Firebase Authentication
- Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Tap the “Sign Up” option	N/A	Registration screen opens	Screen opened	Pass	N/A
2	Enter a valid email	test@gmail.com	Email accepted	Email accepted	Pass	N/A
3	Enter a valid password	*****	Password meets criteria	Password accepted	Pass	N/A
5	Tap “Create Account”	N/A	Account created	Account created	Pass	User registered
6	Redirect to dashboard	N/A	Dashboard displayed	Dashboard displayed	Pass	Auto login

Post Condition: A new user account is created, and the user is logged into the NutriTrack dashboard.

Test Case ID: NT-TC-02

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): User Login

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

This test case verifies that a registered user can successfully log in to the NutriTrack application using valid credentials on an iOS device.

Context: Secure authentication is required to allow users to access their personal nutrition data and application features.

Precondition

- User has a registered NutriTrack account
- The NutriTrack app is installed on the iPhone
- A stable internet connection is available

Dependency

- Firebase Authentication
- iOS network services

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Launch the NutriTrack app	N/A	The login screen is displayed	Login screen displayed	Pass	N/A
2	Enter a valid email	test@gmail.com	Email is accepted	Email accepted	Pass	N/A
3	Enter a valid password	*****	Password is accepted	Password accepted	Pass	N/A
4	Tap the Login button	N/A	User is redirected to the dashboard	Dashboard displayed	Pass	Successful login

Post Condition: User is successfully logged in and redirected to the dashboard screen.

Test Case ID: NT-TC-03

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Forgot Password Functionality

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

This test case verifies that users can reset their password using the “Forgot Password” feature by receiving a password reset link via email.

Context: The Forgot Password feature allows users to securely regain access to their accounts if they forget their login credentials.

Precondition

- User has a registered email address
- The NutriTrack app is installed on an iOS device
- An active internet connection is available

Dependency

- Firebase Authentication
- Email delivery service

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Tap the “Forgot Password” link	N/A	Password reset screen opens	Screen opened	Pass	N/A
2	Enter registered email	test@gmail.com	Email accepted	Email accepted	Pass	N/A
3	Tap “Send Reset Link”	N/A	The reset email has been sent	Email received	Pass	Firebase email sent
4	Open the email and click the reset link	N/A	Redirected to password reset page	Redirected successfully	Pass	External browser opened

Post Condition: User receives a password reset email and can successfully proceed to set a new password.

Test Case ID: NT-TC-04
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): Nutrition Dashboard Display
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)
Description: This test case verifies that the dashboard correctly displays daily calorie and macronutrient intake.

Context: The dashboard provides users with a visual overview of their daily nutrition progress.

Precondition

- User is logged in
- At least one meal is logged

Dependency

- Firestore Database
- Data processing logic

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Navigate to the dashboard	N/A	Dashboard loads	Loaded successfully	Pass	N/A
2	View calorie summary	N/A	Correct calorie total shown	Correct total shown	Pass	N/A
3	View macro breakdown	N/A	Macros displayed visually	Displayed correctly	Pass	Charts visible

Post Condition:
The user can view accurate daily nutritional insights on the dashboard.

Test Case ID: NT-TC-05

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Manual Meal Logging

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description: This test case verifies that users can manually add meal details and view the logged meal in the daily summary.

Context: Manual meal logging allows users to record meals.

Precondition

- User is logged in
- The dashboard screen is accessible

Dependency

- Firebase Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Navigate to Custom Meal	N/A	Meal entry screen opens	Screen opened	Pass	N/A
2	Enter food name	Rice & Vegetables	Food name accepted	Accepted	Pass	N/A
3	Enter macros	450 kcal	Macros saved	Saved correctly	Pass	N/A
4	Save meal	N/A	Meal appears in the daily log	Meal displayed	Pass	Logged successfully

Post Condition:

Meal data is saved and reflected in the daily nutrition summary.

Test Case ID: NT-TC-06
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): AI Meal Planner Suggestions
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)
Description: This test case verifies that the AI Meal Planner generates personalised meal suggestions based on the user’s nutrition goals and dietary preferences.

Context: The AI Meal Planner enhances user experience by providing intelligent meal recommendations aligned with calorie and macronutrient targets.

Precondition

- User is logged in
- Nutrition goals (calories and macros) are already set
- An active internet connection is available

Dependency

- AI Meal Planning API
- Firestore Database
- Backend recommendation logic

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Navigate to AI Meal Planner	N/A	The AI planner screen opens	Screen opened	Pass	N/A
2	Enter Meal Preferences and tap “Create Meals”	High Protein wrap	System processes the request	Processing started	Pass	Loading indicator shown
4	Display meal suggestions	N/A	Personalised meals displayed	Meals displayed	Pass	Suggestions shown
5	View nutritional details	Suggested meal	Calories & macros shown	Displayed correctly	Pass	Accurate values

Post-Condition: The user receives AI-generated meal suggestions aligned with their nutrition goals and preferences.

Test Case ID: NT-TC-07

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Add Food in My Foods (AI Meal Planner & Custom Food)

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description: This test case verifies that when a user taps the plus button, they are redirected to the AI Meal Planner page, where they can either save a generated AI-based meal suggestion or a custom food.

Context: The plus button serves as a centralised entry point for food addition, enabling users to choose between AI-generated meals or manually enter their own food for flexibility and efficiency.

Precondition

- The dashboard screen is accessible
- An internet connection is available

Dependency

- AI Meal Planner API
- Firestore Database
- Backend business logic

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	View AI meal suggestions	N/A	Meals displayed with nutrition	Meals displayed	Pass	Suggestions visible
2	Save selected meal	Suggested meal	Meal saved to daily log	Meal saved	Pass	Stored in the database
3	Tap “Add Custom Food”	N/A	Custom food form opens	Form opened	Pass	Alternative option
4	Enter food details	Food name & calories	Data accepted	Accepted	Pass	Manual entry
5	Save custom food	N/A	Food saved successfully	Food saved	Pass	Appears in My Foods

Post Condition:

User successfully adds food either by saving an AI-generated meal or by manually entering a custom food item.

Test Case ID: NT-TC-08

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Edit Food in My Foods

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description: This test case verifies that users can edit the details of an existing food item in the “My Foods” page and save the updated information successfully.

Context: The edit functionality allows users to correct or update the nutritional information of saved foods, ensuring accuracy and personalisation.

Precondition

- At least one food item exists in the “My Foods” list
- An internet connection is available

Dependency

- Firestore Database
- User authentication state

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Navigate to the “My Foods” page	N/A	My Foods list is displayed	List displayed	Pass	N/A
2	Select an existing food item	Boiled Eggs	Food details page opens	Opened successfully	Pass	N/A
3	Tap the “Edit” option	N/A	The editable form is displayed	Form opened	Pass	N/A
4	Update the food details	N/A	Details updated	Updated correctly	Pass	N/A
7	Save changes	N/A	Changes saved successfully	Saved successfully	Pass	Database updated
8	Return to My Foods list	N/A	Updated food details are visible	Updated data shown	Pass	Persisted

Post Condition:

The selected food item is updated, and the new information is reflected in the “My Foods” list.

Test Case ID: NT-TC-09

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Delete Food from My Foods

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

This test case verifies that users can delete a previously saved food item from the “My Foods” page.

Context

Deleting foods allows users to manage and clean up incorrect or unwanted food entries.

Precondition

- User is logged in
- At least one food item exists in “My Foods”

Dependency

- Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open My Foods page	N/A	Food list displayed	Displayed	Pass	N/A
2	Select a food item	Boiled Eggs	Food details open	Opened	Pass	N/A
3	Tap the Delete button	N/A	Confirmation dialogue shown	Shown	Pass	N/A
4	Confirm delete	N/A	Food removed from the list	Removed	Pass	Database updated

Post Condition:

The food item is permanently deleted from My Foods.

Test Case ID: NT-TC-10

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Edit Today's Logged Meal Entry

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description: This test case verifies that a user can edit a meal entry logged for the current day and that the daily nutrition totals update correctly.

Context: Editing a logged meal helps users correct mistakes and maintain accurate daily nutrition tracking.

Precondition

- At least one meal is logged for today
- An internet connection is available

Dependency

- Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Today's Log	N/A	Today's entries displayed	Displayed	Pass	N/A
2	Select an existing meal entry	Momo	Meal details open	Opened	Pass	N/A
3	Tap Edit	N/A	Edit form shown	Shown	Pass	N/A
4	Update macros	350 kcal	Values updated	Updated	Pass	N/A
5	Save changes	N/A	Entry updated and saved	Saved	Pass	Firestore updated
6	Return to dashboard	N/A	Daily totals updated	Updated	Pass	Verified

Post Condition:

Meal entry reflects updated values and dashboard totals recalculate correctly.

Test Case ID: NT-TC-11

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Delete Today's Logged Food (Settings Page)

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

This test case verifies that users can delete a food entry logged for the current day via the Settings page.

Context

This feature allows users to correct accidental or incorrect daily food entries.

Precondition

- User is logged in
- At least one food entry exists for today

Dependency

- Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open the Settings page	N/A	Settings displayed	Displayed	Pass	N/A
2	Open Today's Entries	N/A	Entries listed	Listed	Pass	N/A
3	Select food entry	Logged meal	Entry selected	Selected	Pass	N/A
4	Delete entry	N/A	Entry removed	Removed	Pass	Totals updated

Post Condition:

Daily nutrition summary updates correctly.

Test Case ID: NT-TC-12

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Change Password

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

This test case verifies that users can securely change their account passwords.

Context

Password change functionality improves account security and user control.

Precondition

- User is logged in

Dependency

- Firebase Authentication

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Settings	N/A	Settings page shown	Shown	Pass	N/A
2	Tap Change Password	N/A	Password form opens	Opened	Pass	N/A
3	Enter new password	*****	Password accepted	Accepted	Pass	N/A
4	Save password	N/A	Password updated	Updated	Pass	Firebase sync

Post Condition:

The user can log in using the new password.

Test Case ID: NT-TC-13
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): Update User Profile
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:
This test case verifies that users can update personal profile information.

Context: Profile updates allow personalisation and accurate user data storage.

Precondition

- User is logged in

Dependency

- Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Profile page	N/A	Profile displayed	Displayed	Pass	N/A
2	Edit profile fields	Name, age	Fields editable	Editable	Pass	N/A
3	Save changes	N/A	Profile updated	Updated	Pass	Stored

Post Condition:
Updated profile information is saved successfully.

Test Case ID: NT-TC-14

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Update Macro Goals

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

This test case verifies that users can update calorie and macronutrient goals.

Context: Goal updates enable users to adjust their nutrition targets in response to lifestyle changes.

Precondition

- User is logged in

Dependency

- Firestore Database

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open the Goals page	N/A	Goals displayed	Displayed	Pass	N/A
2	Edit macro values	Protein, carbs	Values accepted	Accepted	Pass	N/A
3	Save goals	N/A	Goals updated	Updated	Pass	Reflected

Post Condition:

New macro goals are applied to calculations.

Test Case ID: NT-TC-15

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Search Food in My Foods List

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description: This test case verifies that the search function in “My Foods” returns correct matching items based on user input.

Context: Searching improves efficiency for users with many saved foods by reducing scrolling time.

Precondition

- User is logged in
- Multiple food items exist in “My Foods”

Dependency

- Local filtering/search logic
- Firestore Database (data retrieval)

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open My Foods page	N/A	Food list displayed	Displayed	Pass	N/A
2	Tap the search bar	N/A	Keyboard opens	Opened	Pass	N/A
3	Type food keyword	“Egg”	Matching items shown	Shown	Pass	Filter correct
4	Clear search	N/A	Full list returns	Returned	Pass	N/A

Post Condition:

The user can quickly find saved foods using search.

Test Case ID: NT-TC-16
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): Logout Functionality
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:
This test case verifies that users can log out of the application.

Context

Logout ensures session security on shared devices.

Precondition

● User is logged in

Dependency

● Firebase Authentication

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Settings	N/A	Settings shown	Shown	Pass	N/A
2	Tap Logout	N/A	Redirect to login	Redirected	Pass	Session cleared

Post Condition:
User session ends successfully.

Test Case ID: NT-TC-17
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): Internet Failure Handling
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:
Verifies app behaviour when the internet is unavailable.

Context

Ensures graceful error handling.

Precondition

• Internet disabled

Dependency

• Network state

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Disable internet	N/A	App detects issue	Detected	Pass	N/A
2	Use the AI feature	N/A	Error message shown	Shown	Pass	User informed

Post Condition:
App remains stable.

Test Case ID: NT-TC-18

Tested By: Tenzin Kunsel Sherpa

Module Name (Test Title): Account Deletion

Test Executed Date: 28/12/2025

Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:

Verifies that users can permanently delete their accounts.

Context

Supports user data control and privacy.

Precondition

- User logged in

Dependency

- Firebase Authentication
- Firestore

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Settings	N/A	Settings shown	Shown	Pass	N/A
2	Delete account	N/A	Account removed	Removed	Pass	Data erased

Post Condition:

User account is permanently deleted.

Test Case ID: NT-TC-19
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): App Launch Performance (iOS)
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:
Verifies app launches successfully on iOS devices.

Context: Performance impacts usability.

Precondition

- App installed

Dependency

- Device hardware

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Launch app	N/A	App opens < 3 seconds	Opened	Pass	Acceptable

Post Condition:
Application launches successfully.

Test Case ID: NT-TC-20
Tested By: Tenzin Kunsel Sherpa
Module Name (Test Title): Persist Login Session After App Restart
Test Executed Date: 28/12/2025
Test Executed Location: iOS Mobile Application (iPhone – iOS)

Description:
This test case verifies that the user remains logged in after closing and reopening the NutriTrack app, ensuring session persistence.

Context: Session persistence improves usability by preventing users from needing to log in repeatedly.

Precondition

- User is logged in successfully
- The device has a stable internet connection

Dependency

- Firebase Authentication session handling
- Local secure storage (iOS)

Test Steps

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Confirm user is logged in	N/A	Dashboard visible	Visible	Pass	N/A
2	Close the app fully	N/A	App closes	Closed	Pass	Swipe up close
3	Reopen NutriTrack	N/A	User remains logged in	Logged in	Pass	No login prompt
4	Navigate to the dashboard	N/A	Dashboard loads normally	Loaded	Pass	Session retained

Post Condition:
User stays authenticated after app restart without re-entering credentials.